Of Housewives and Feminists as a Game — class version

Lecture 3 case. Short version of `Cases/Housewive and feminists/housewives-game.pdf`, where every derivation is worked out. Paper: Görges (2021), *Labour Economics* 72, 102044; "p. N" = article page. Amounts in ECU (1 ECU = 1 euro cent). [Ours] marks our analysis, not the paper's.

(2) The game

Real couples (192 subjects) play a one-shot battle of the sexes, simultaneously and without communication, once with the partner and once with a stranger (pp. 2–4). **Players:** woman W, man M. **Actions:** Career (C) or Family (F); in the Neutral frame they are A and B and the payoffs are identical. **Payoffs** (Table 1, p. 3, $\alpha = 200$, $\beta = 100$): the Career player against a Family partner gets 200, the Family player 100, mis-coordination 0.

		Man C	**Man** F
Woman	C	0, 0	**200**, **100**
	F	**100**, **200**	0, 0

(3) Equilibria

Pure NE: (C, F) (Progressive, female-favouring) and (F, C) (Traditional, male-favouring); total payoff 300.

Mixed NE (principle of indifference). Let $q = \Pr(M \text{ plays C})$. The woman gets $u_W(\mathrm{C}, q) = 200(1-q)$ and $u_W(\mathrm{F}, q) = 100\,q$; she is indifferent iff $q = \frac{200}{300} = \frac{2}{3}$, and prefers C iff $q < \frac{2}{3}$. By symmetry $p = \Pr(W \text{ plays C}) = \frac{2}{3}$. Each player's mix is pinned down by the *other's* payoffs.

Mixed NE: $\Pr(\mathrm{C}) = \frac{\alpha}{\alpha+\beta} = \frac{2}{3}$ each; $u = \frac{\alpha\beta}{\alpha+\beta} = \frac{200}{3} \approx 66.7$ (p. 3). Outcomes: Progressive $\frac{2}{9}$, Traditional $\frac{2}{9}$, both-Career $\frac{4}{9}$, both-Family $\frac{1}{9}$, so coordination is $\frac{4}{9} \approx 44.4\%$ ("just below 45%", p. 3). Total payoff $2 \times \frac{200}{3} \approx 133.3$, against 300 in a pure NE: the mixed NE is equal but inefficient, the pure NE efficient but unequal (200 against 100).

(4) Focal points

Schelling (1960): a salient label can select an equilibrium without changing payoffs, by making one outcome the common expectation. The paper's hypothesis (p. 3): the Career/Family label makes the Traditional equilibrium focal, so it is reached more often, "improving efficiency at the cost of increased inequality to the disadvantage of women". In game terms the label shifts beliefs: a woman who expects the man to play C with probability above $\frac{2}{3}$ best-responds with F. This is the efficiency–equality trade-off: total payoff moves toward 300, but the man gets 200 and the woman 100.

(5) Theory against the data

Career rate (Table 4 col. 1, p. 8)	Men Neutral	Women Neutral	Men Norm	Women Norm
Observed	0.536	0.563	0.625	0.250

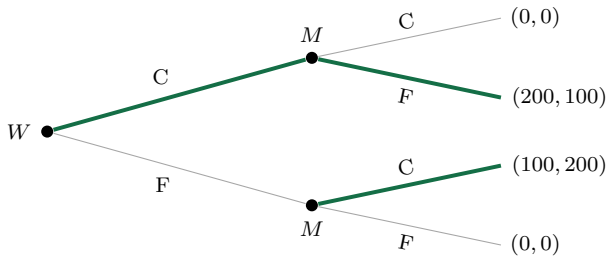
Outcomes (Table 3 col. 2, p. 6): Norm versus Neutral changes Progressive by −0.171, Traditional by +0.255, non-coordination by −0.085. **Earnings** (Table C1, p. 15): Norm men 112.50, women 75.00 (women earn $\frac{2}{3}$, "67% of men", p. 7); Neutral men 80.36, women 83.04.

[Ours] **Best responses against the real rates.** A woman in Norm facing men's 0.625: C pays $0.375 \times 200 = 75$, F pays $0.625 \times 100 = 62.5$, so Career was her best response, yet only 25% chose it. A man facing women's 0.25: C pays 150, F pays 25. Women's beliefs about men overshoot the true rates (fn. 27, p. 9); with a belief above $\frac{2}{3}$, F is the woman's best response.

[Ours] **Independence benchmark** (*ours, not the paper's*): with the observed rates and independent choices, Neutral gives Progressive 0.261, Traditional 0.234, coordination 0.495; Norm gives 0.094, 0.469 and 0.5625. Expected Norm earnings: men 103.1, women 65.6, below the observed 112.50 and 75.00, so pairs coordinate more than independent play implies. Total earnings per couple rise from 163.4 to 187.5 with the Norm label, but men gain 32.14 and women lose 8.04. **Own-payoff puzzle** (ours): an identity utility n for women choosing Family would move the *men's* mix, $\frac{200-n}{300}$, not the women's; the data move the women, which points to equilibrium selection (a focal point) rather than a payoff shift.

(8) Dynamics: the sequential BoS

If the woman moves first and the man observes her move, backward induction gives: after C he plays F ($100 > 0$); after F he plays C ($200 > 0$); she compares C → 200 with F → 100 and plays C.



SPE: W plays C; M plays F after C and C after F; payoffs $(200, 100)$. The first mover gets 200, against $\frac{200}{3} \approx 66.7$ in the mixed NE, a gain of $\frac{400}{3} \approx 133.3$. There is also a Nash equilibrium that is not subgame perfect, $(\text{F};\ \text{C}, \text{C})$ with payoffs $(100, 200)$, resting on the threat "I play Career whatever you do": after C the man would get 0 by carrying it out and 100 from F, so it is not credible. **Commitment** is valuable: whoever commits first (job offer, first child) is the one the other must best-respond to. **Path dependence** (fn. 12, p. 4): today's specialisation raises the payoff of repeating it tomorrow. **[Ours] Lecture 5 preview:** alternating $(\text{C}, \text{F}), (\text{F}, \text{C}), \ldots$ gives 150 per period each against $\frac{200}{3}$ for repeating the mixed NE.

Takeaway

- The one-shot BoS has two efficient but unequal pure equilibria and an inefficient mixed one ($\Pr(\text{C}) = \frac{2}{3}$, payoff $\frac{200}{3}$, coordination $\frac{4}{9}$); theory alone does not say which is played.
- The Norm label acts as a focal point: women's Career rate falls from 0.563 to 0.250, the Traditional outcome rises by 0.255 and women earn $\frac{2}{3}$ of men's earnings, with no change in payoffs.
- Moving first is worth 133.3 ECU; the credible commitment, not the threat, decides who specialises in Career.